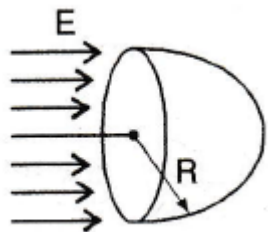


NSG ELECTROSTATS

Physics

Section A

1. According to the figure, a hemispherical object is placed in an electric field, find the outward flux through its curved surface:



- A) $E\pi R^2$ B) $\frac{E\pi R^2}{2}$ C) $E2\pi R$ D) $E(\pi R)$

Answer: A,

Solution:

Total outward flux

$$\phi = \phi_{cs} = \phi_n = \text{flux through the curved surface}$$

$$\text{Effective area} = \pi R^2$$

$$\phi = E\pi R^2 \cos 180^\circ$$

$$|\phi| = +E\pi R^2$$

2. A cylinder of radius R and length L is placed in a uniform electric field E parallel to the cylinder axis. The total flux for the surface of the cylinder is given by:

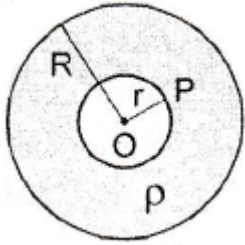
- A) $2\pi R^2 E$ B) $\pi R^2 / E$ C) $\frac{\pi R^2 + \pi R^2}{E}$ D) Zero

Answer: D,

Solution:

$$-\pi R^2 E + \pi R^2 E = 0$$

3. A insulated sphere of radius R has a uniform volume charge density ρ . The electric field at a point P inside the sphere and distant r from the centre is:



- A) $\frac{R\rho}{3\varepsilon_0}$ B) $\frac{r\rho}{3\varepsilon_0}$ C) zero D) $\left(\frac{r\rho}{\varepsilon_0}\right)$

Answer: B,

Solution:

Apply Gauss's law

$$\oint \vec{E} \cdot d\vec{s} = \frac{\text{Charge in closed}}{\varepsilon_0}$$

$$E4\pi r^2 = \frac{\frac{4}{3}\pi r^3 \rho}{\varepsilon_0} \text{ or } E = \frac{\rho r}{3\varepsilon_0}$$

4. If the electric flux entering and leaving an enclosed surface respectively is ϕ_1 and ϕ_2 , the electric charge inside the surface will be_____

- A) $(\phi_2 - \phi_1)\varepsilon_0$ B) $(\phi_1 + \phi_2)/\varepsilon_0$ C) $(\phi_2 - \phi_1)/\varepsilon_0$ D) $(\phi_1 + \phi_2)\varepsilon_0$

Answer: A,

Solution:

The electric flux through enclosed surface is $\phi = \frac{1}{\varepsilon_0} \Sigma q$

$$\therefore \phi = \phi_2 - \phi_1 = \frac{Q}{\varepsilon_0}$$

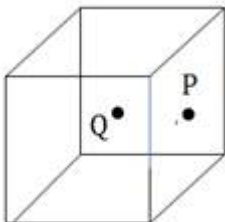
$$\Rightarrow Q = (\phi_2 - \phi_1)\varepsilon_0$$

5. A charge Q is situated at the centre of a cube. The electric flux through one of the faces of the cube is_____

- A) Q/ε_0 B) $Q/2\varepsilon_0$ C) $Q/4\varepsilon_0$ D) $Q/6\varepsilon_0$

Answer: D,

Solution:



Total Electric flux through cube is

$$\phi = \frac{q_{enc}}{\epsilon_0}$$

$$\therefore \phi = \frac{Q}{\epsilon_0}$$

since there are 6 faces

flux through each side will be

$$\therefore \phi_{side} = \frac{Q}{6\epsilon_0}$$

6. In a region of space, the electric field is given by $\vec{E} = 8\hat{i} + 4\hat{j} + 3\hat{k}$. The electric flux through a surface of area 100 units in the xy plane is

- A) 800 units B) 300 units C) 400 units D) 1500 units

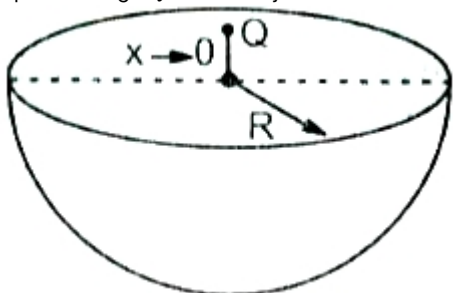
Answer: B,

Solution:

$$\vec{E} = 8\hat{i} + 4\hat{j} + 3\hat{k}, \vec{A} = 100\hat{k}$$

$$\text{So } \phi = \vec{E} \cdot \vec{A} = 300 \text{ units}$$

7. A point charge Q is located just above the centre of the flat face of hemisphere as shown in figure. The electric flux through the flat face and curved face of hemisphere are respectively,



- A) $\frac{Q}{2\epsilon_0}, -\frac{Q}{2\epsilon_0}$ B) $-\frac{Q}{2\epsilon_0}, \frac{Q}{2\epsilon_0}$ C) $\frac{Q}{\epsilon_0}, -\frac{Q}{\epsilon_0}$ D) $-\frac{Q}{\epsilon_0}, \frac{Q}{\epsilon_0}$

Answer: B,

Solution:

Place another identical hemisphere on given one

so that flat faces of two coincide. Then total flux

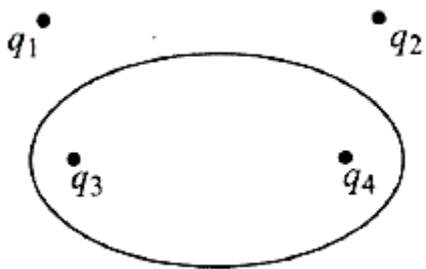
linked with cross-section of bottom half (lower

hemisphere) is $\frac{Q}{2\epsilon_0}$, as charge enclosed by given

hemisphere is zero, so flux linked with flat face

of hemisphere is $\frac{-Q}{2\epsilon_0}$

8. Fig. 2.95 shows four charges q_1 , q_2 , q_3 , and q_4 fixed in space. Then the total flux of the electric field through a closed surface S , due to all the charges, is



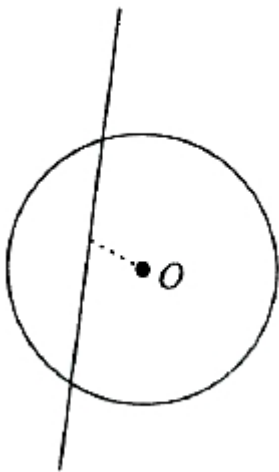
- A) not equal to the total flux through S due to q_3 and q_4 B) equal to the total flux through S due to q_3 and q_4 C) zero if $q_1 + q_2 = q_3 + q_4$ D) twice the total flux through S due to q_3 and q_4 if $q_1 + q_2 = q_3 + q_4$

Answer: B,

Solution:

Net flux is due to charges inside S only.

9. A uniformly charged and infinitely long line having a linear charge density λ is placed at a normal distance y from a point O . Consider a sphere of radius R with O as the center and $R > y$. Electric flux through the surface of the sphere is



- A) zero B) $\frac{2\lambda R}{\epsilon_0}$ C) $\frac{2\lambda\sqrt{R^2 - y^2}}{\epsilon_0}$ D) $\frac{\lambda\sqrt{R^2 + y^2}}{\epsilon_0}$

Answer: C,

Solution:

$$\text{Electric flux } \oint_S \vec{E} \cdot d\vec{s} = \frac{q_{\text{in}}}{\epsilon_0}$$

q_{in} is the charge enclosed by the Gaussian surface, which, in the present case, is the surface of the given sphere. As shown, in length AB of line lies inside the sphere. In $\triangle OO'A$,

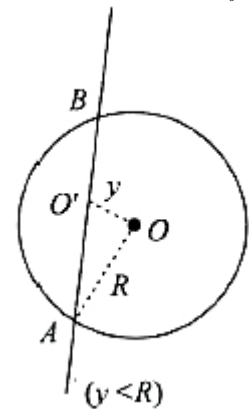
$$R^2 = Y^2 + (O'A)^2$$

$\therefore$
and
Charge on length AB is $2\sqrt{R^2 - y^2} \times \lambda$
Therefore, electric flux is

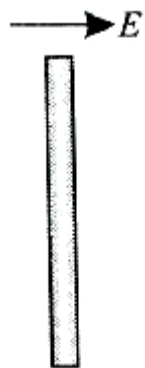
$$O'A = \sqrt{R^2 - y^2}$$

$$AB = 2\sqrt{R^2 - y^2}$$

$$\oint_S \vec{E} \cdot d\vec{s} = \frac{2\lambda\sqrt{R^2 - y^2}}{\epsilon_0}$$



10. An uncharged conducting large plate is placed as shown. Now an electric field E toward right is applied. Find the induced charge density on the right surface of the plate.



- A) $-\epsilon_0 E$ B) $\epsilon_0 E$ C) $-2\epsilon_0 E$ D) $2\epsilon_0 E$

Answer: B,

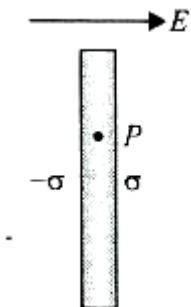
Solution:

Net electric field at point P should be zero.

For this electric field due to induced charges

is equal to the applied electric field. So

$$\frac{\sigma}{\epsilon_0} = E \text{ or } \sigma = \epsilon_0 E$$



11. At a point in space, the electric field points towards north. In the region surrounding this point, the rate of change of potential will be zero along
A) north B) south C) north-south D) east-west

Answer: D,

Solution:

Equipotential surfaces are perpendicular to the electric lines of forces.

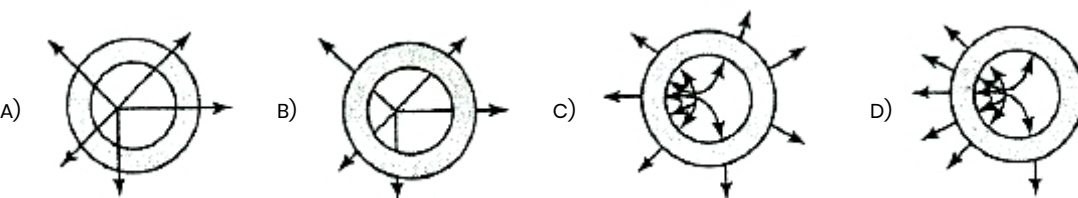
12. The electric flux from a cube of edge l is ϕ . If an edge of the cube is made $2l$ and the charge enclosed is halved, its value will be
A) 4ϕ B) 2ϕ C) $\phi/2$ D) ϕ

Answer: C,

Solution:

$$\phi = \frac{q}{\epsilon_0} \text{ (Independent of dimensions)}$$

13. A metallic shell has a point charge q kept inside its cavity. Which of the following diagrams correctly represents the electric lines of forces?

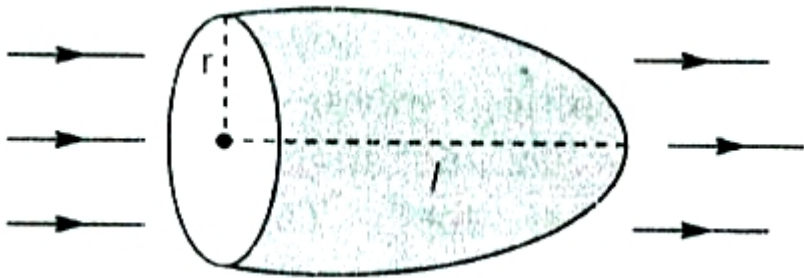


Answer: C,

Solution:

Electric field is perpendicular to the equipotential surfaces and is zero everywhere inside the metal.

14. Fig. shows a circular surface and a paraboloidal surface. It is placed in a uniform electric field of magnitude E such that the circular surface is oriented at right-angles to the direction of field. Electric flux through the paraboloidal surface is:



- A) zero B) $\pi r^2 l E$ C) $\frac{1}{2} \pi r^2 E$ D) $\pi r^2 E$

Answer: D,

Solution:

Flux through the circular surface, since the field is uniform, will be

$$\phi = \vec{E} \cdot \vec{S}$$

$$= ES \cos 180^\circ$$

$$= -E\pi r^2 [\vec{S} \text{ being outward normal to surface is opposite to } \vec{E}]$$

All the lines of force that enter at the circular opening also leave through the paraboloidal surface, so that, flux through the paraboloidal surface has the same magnitude, but opposite sign.

$$\text{i.e., } \phi = E\pi r^2$$

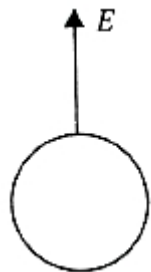
15. A cylinder of radius R and length l is placed in a uniform electric field E parallel to the axis of the cylinder. The total flux over the curved surface of the cylinder is

- A) zero B) $\pi R^2 E$ C) $2\pi R^2 E$ D) $E/\pi R^2$

Answer: A,

Solution:

At any point on the curved surface, E and area vector are perpendicular to each other.



16. A system consists of a thin charged wire ring of radius r and a very long uniformly charged wire oriented along the axis of the ring, with one of its ends coinciding with the center of the ring. The total charge on the ring is q , and the linear charge density on the straight wire is λ . The interaction force between the ring and the wire is

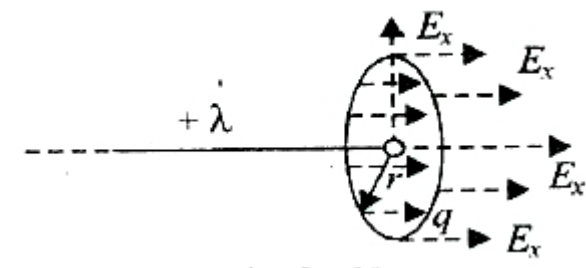
A) $\frac{\lambda q}{4\pi\epsilon_0 r}$ B) $\frac{\lambda q}{2\sqrt{2}\pi\epsilon_0 r}$ C) $\frac{2\sqrt{2}\lambda q}{\pi\epsilon_0 r}$ D) $\frac{4\lambda q}{\pi\epsilon_0 r}$

Answer: A,

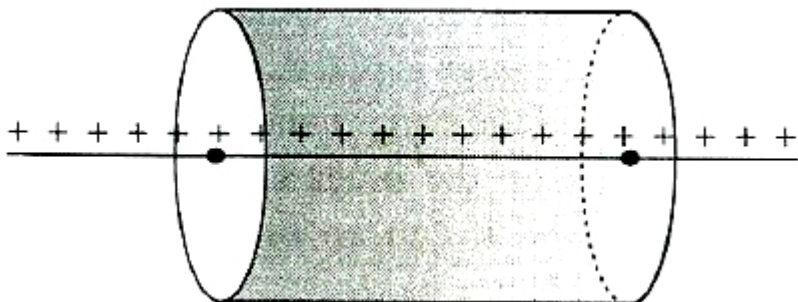
Solution:

Net force $F_{\text{net}} = qE_x$

$$F = q \frac{\lambda}{4\pi\epsilon_0 r} = \frac{\lambda q}{4\pi\epsilon_0 r}$$



17. Consider an infinite line charge having uniform linear charge density and passing through the axis of a cylinder. What will be the effect on the flux passing through the curved surface if the portions of the line charge outside the cylinder are removed.



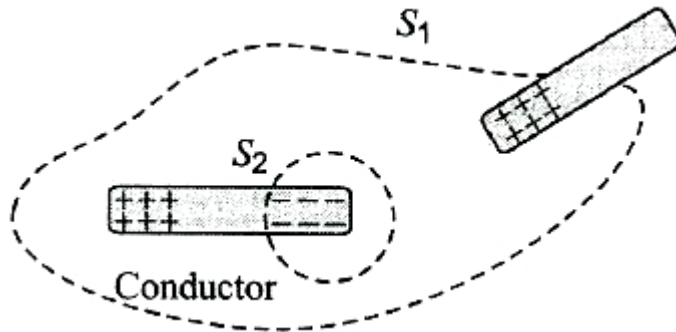
- A) decreases B) increases C) remains same D) cannot say

Answer: A,

Solution:

If portion outside the cylinder is removed, electric field at points on curved surface will decrease.

18. Charge on an originally uncharged conductor is separated by holding a positively charged rod very closely nearby, as shown in



. Assume that the induced negative charge on the conductor is equal to the positive charge q on the rod. Then the flux through surface S_1 is

- A) zero B) q/ϵ_0 C) $-q/\epsilon_0$ D) none of these

Answer: B,

Solution:

Net charge on the conductor will be zero. So, net charge inside

S_1 will be the charge on the rod. Hence, flux through S_1 is q/ϵ_0 .

19. A cube of side 10 cm encloses a charge of $0.1\mu\text{C}$ at its center. Calculate the number of lines of force through each face of the cube.

- A) 1.113×10^{11} B) 1.13×10^4 C) 1.13×10^9 D) 1883

Answer: D,

Solution:

$$\frac{q}{6\epsilon_0} = \frac{0.1 \times 10^{-5}}{6 \times 8.85 \times 10^{-12}} = 1883$$

20. Consider two concentric spherical surfaces S_1 with radius a and S_2 with radius $2a$, both centered at the origin. There is a charge $+q$ at the origin and there are no other charges. Compare the flux ϕ_1 through S_1 with the flux ϕ_2 through S_2 .

- A) $\phi_1 = 4\phi_2$ B) $\phi_1 = 2\phi_2$ C) $\phi_1 = \phi_2$ D) $\phi_1 = \phi_2/2$

Answer: C,

Solution:

Flux through both will be same as the net charge enclosed by both is same.

21. A charge q is placed at the centre of the open end of a cylindrical vessel as shown in fig. The flux of the electric field through the surface of the vessel is

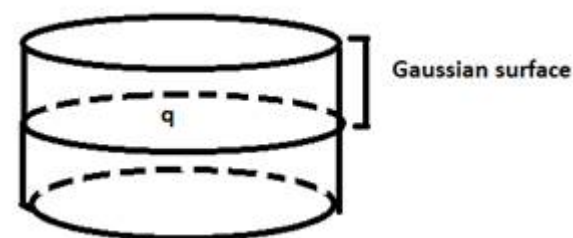


- A) $4q/\epsilon_0$ B) q/ϵ_0 C) $q/2\epsilon_0$ D) $2q/\epsilon_0$

Answer: C,

Solution:

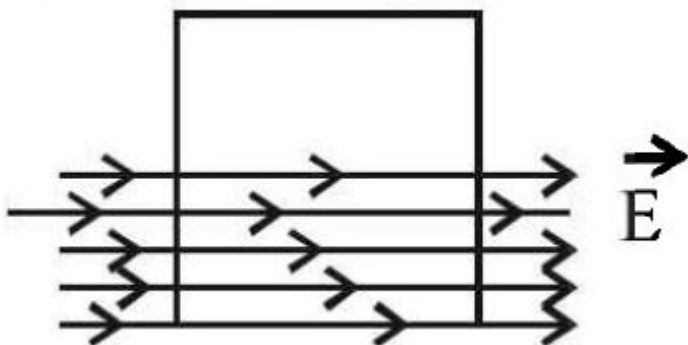
According to Gauss's Law, the flux through a closed cylindrical Gaussian surface is q/ϵ_0 . But the question is about an open cylindrical vessel. Now, take another identical vessel and make a closed Gaussian surface enclosing the charge, as shown in the following figure.



Now, the charge enclosed by two of the surfaces is q . Hence, the charge enclosed by only of the surfaces will be $q/2$

Therefore, the net flux will be half of the total = $q/2\epsilon_0$

22. A square surface of side L meters is in the plane of the paper. A uniform electric field $\vec{E}$ (volt/m), also in the plane of the paper, is limited only to the lower half of the square surface (see figure). The electric flux in SI units associated with the surface is



- A) $EL^2/2$ B) Zero C) EL^2 D) $EL^2/(2\epsilon_0)$

Answer: B,

Solution:

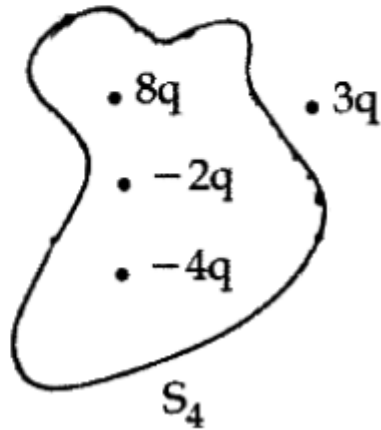
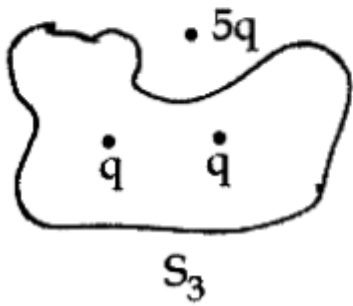
$$\text{Flux} = \vec{E} \cdot \vec{A}$$

$\vec{E}$ is electric field vector & $\vec{A}$ is area vector.

Here, angle between $\vec{E}$ & $\vec{A}$ is 90° .

So, $\vec{E} \cdot \vec{A} = 0$; Flux = 0

23. Four closed surfaces and corresponding charge distributions are shown below –



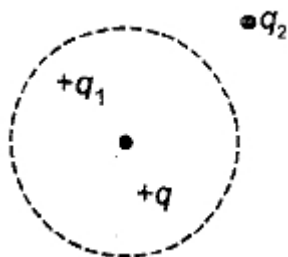
Let the respective electric fluxes through the surfaces be Φ_1 , Φ_2 , Φ_3 and Φ_4 Then –

- A) $\Phi_1 > \Phi_2 > \Phi_3 > \Phi_4$ B) $\Phi_1 < \Phi_2 = \Phi_3 > \Phi_4$ C) $\Phi_1 > \Phi_3; \Phi_2 > \Phi_4$ D) $\Phi_1 = \Phi_2 = \Phi_3 = \Phi_4$

Answer: D,**Solution:**

q = same in all Φ = Same

24. Consider the charge configuration and a spherical gaussian surface as shown in the figure. When calculating the flux of the electric field over the spherical surface, the electric field will be due to



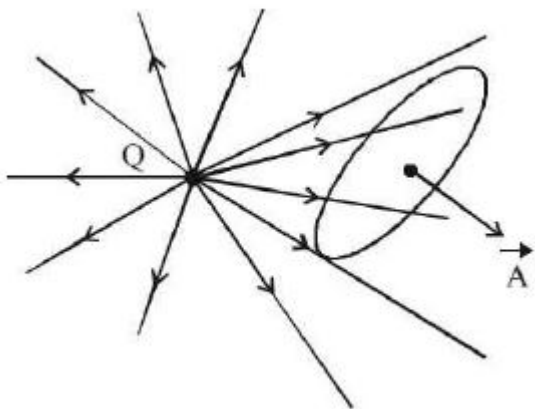
- A) q_2 B) only the positive charges C) all the charges D) $+q_1$ and $-q_1$

Answer: C,

Solution:

Electric flux may be due to the charges present inside the Gaussian surface, but for the purpose of calculation of electric field E at any point, we shall have to consider the contribution of all the charges.

25. In the figure, the net electric flux through the area A is $\phi = \vec{E} \cdot \vec{A}$ when the system is in air. On immersing the system in water the net electric flux through the area



- A) Becomes zero B) Remains same C) Increases D) Decreases

Answer: D,

Solution:

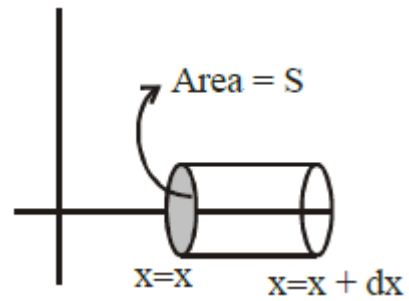
Decreases inside water, therefore flux $\phi = \vec{E} \cdot \vec{A}$ also decreases.

26. The electric potential in a certain region of space depends only on x coordinate as $V = -\alpha x^3 + \beta$, where α and β are constants. The volume charge distribution of space charge is given as:

- A) $\rho(x) = 3\alpha\epsilon_0 x^2$ B) $\rho(x) = -3\alpha\epsilon_0 x^2$ C) $\rho(x) = 6\alpha\epsilon_0 x^2$ D) $\rho(x) = 6\alpha\epsilon_0 x$

Answer: D,

Solution:



$$E = -\frac{dv}{dx} = 3\alpha x^2$$

$$\phi_{\text{in}} = -(3\alpha x^2) s$$

$$\phi_{\text{out}} = \{3\alpha(x + dx)^2\} s = 3\alpha s(2x dx)$$

$$\phi_{\text{net}} = (6\alpha x)(s dx)$$

$$\phi_{\text{net}} = \frac{q}{\epsilon_0} = (6\alpha x)(s dx)$$

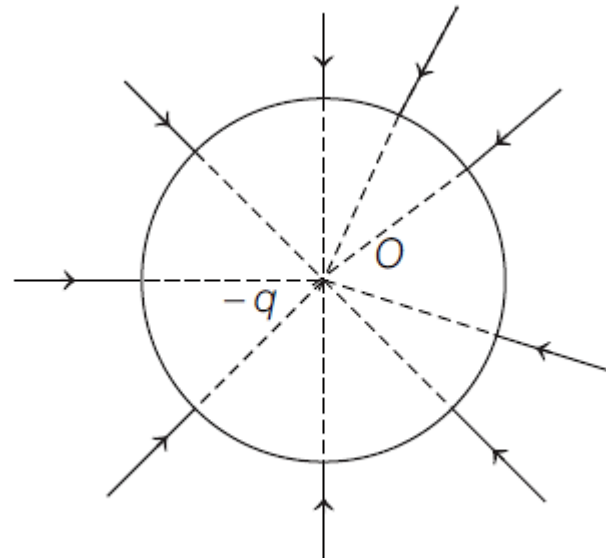
$$\rho = \frac{q}{s dx} = 6\alpha \epsilon_0 x$$

- 27.** A negative charge is placed at the centre of the non-conducting sphere. The direction of the electric field on any point at the surface of the sphere is
- A) Radially inward B) Radially out ward C) Along the tangent to the surface D) No electric field produced

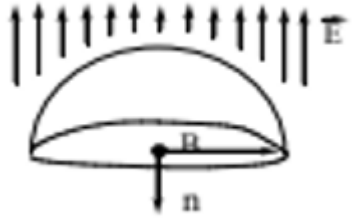
Answer: A,

Solution:

When a negative charge is placed at the centre of the non-conducting sphere. The direction of the electric field on any point at the surface of the sphere is radially inward, as shown in the figure, because electric field lines always enter into the negative charge coming from infinity.



28. If a hemispherical body is placed in a uniform electric field E then the flux linked with the curved surface is



- A) $2\pi R^2 E$ B) $\pi R^2 E$ C) $4\pi R^2 E$ D) $6\pi R^2 E$

Answer: B,

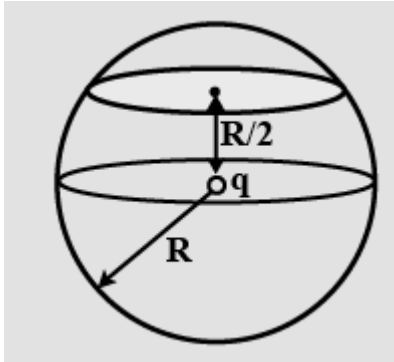
Solution:

Flux passing through the plane surface = the flux passing through the curved surface.

$$\phi = E\pi R^2$$

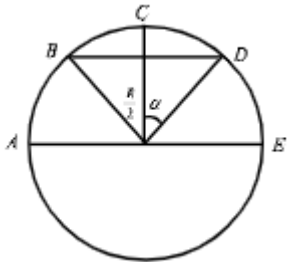
Section B

29. Flux passing through the shaded surface of a sphere is $q/Z\epsilon_0$ when a point charge q is placed at the center. (radius of the sphere is R) Find the value of Z



Answer: 4.00, Rounded Answer: 4

Solution:

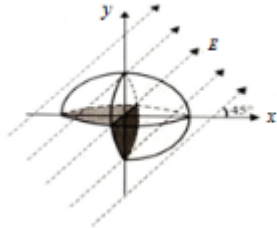


$\alpha = 60^\circ$. Solid angle subtended by BCD is (i) $\omega = 2\pi(1 - \cos \alpha) = \pi$

(ii) Solid angle subtended by $ABDE$ is $\omega_{(ABCDE)} - \omega_{(BCD)} = 2\pi - \pi = \pi$

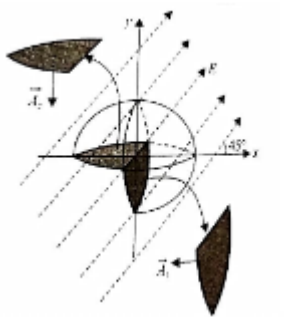
Hence, flux through $ABDE$ is $\phi = \frac{q}{\epsilon_0} \frac{\pi}{4\pi} = \frac{q}{4\epsilon_0}$

30. One-fourth of a sphere of radius $1m$ is removed as shown in the figure. An electric field $\frac{\sqrt{2}}{\pi}$ N/C exists parallel to the xy plane. Find the flux through the remaining curved part is (in Nm^2/C)



Answer: 1.00, Rounded Answer: 1

Solution:



$\phi_{\text{plain}} + \phi_{\text{curve}} = 0$ or $\phi_{\text{plain}} = -\phi_{\text{curve}}$

$$\vec{A}_1 = -\frac{\pi R^2}{2} \hat{i}, \vec{A}_2 = -\frac{\pi R^2}{2} \hat{j}$$

$$\vec{E} = E \cos 45^\circ \hat{i} + E \sin 45^\circ \hat{j}$$

$$\begin{aligned} \phi &= \vec{E} \cdot (\vec{A}_1 + \vec{A}_2) \\ &= \frac{-E \pi R^2}{\sqrt{2} \cdot 2} - \frac{E \pi R^2}{\sqrt{2} \cdot 2} = \frac{-\pi R^2 E}{\sqrt{2}} \end{aligned}$$

This is the flux entering. So flux is $\frac{-\pi R^2 E}{\sqrt{2}}$